SEPARATEDNESS FROM THE DVR VALUATIVE CRITERION OVER A NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4845-4894, following the statement at 4828-4843.

Editorial note. No load-bearing omissions; the codimension-0 preservation step is justified in the text via the etale base change.

Allowed standard prerequisites (editorial).

- Separated implies valuative uniqueness. Morphisms of Spaces, Lemma spaces-morphisms-lemma-separated-implies-valuative (cross-chapter, cited at ~4846-4848). Implication (1) implies (2).
- Properties of the diagonal. Morphisms of Spaces, Lemma spaces-morphisms-lemma-properties-diagonal (cross-chapter, cited at ~4855-4857). Diagonal is representable, separated, a monomorphism, locally of finite type.
- Closed immersion is etale-local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-closed-immersion-local (cross-chapter, cited at ~4860-4862). It suffices to show V -> U is a closed immersion.
- Scheme-level valuative criterion yields properness. Limits, Lemma limits-lemma-Noetherian-dvr-valuative-proper (cross-chapter, cited at ~4889-4890). Concludes V -> U is proper, hence the diagonal is proper.
- Proper monomorphism is a closed immersion. Etale Morphisms, Lemma etale-lemma-characterize-closed-immersion (cross-chapter, cited at ~4891-4893). Concludes the diagonal is a closed immersion.

Source reference. Original equation 0H1V: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4818–4818. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04XV: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H1X: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03XH: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M4: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XG: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZK: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HK: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KU: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04N7: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-separation/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-separated and locally of finite type and Y is locally Noetherian. The following are equivalent:*

1. *The morphism f is separated.*
2. *For any diagram (0H1V) there is at most one dotted arrow.*
3. *For all diagrams (0H1V) with A a discrete valuation ring there is at most one dotted arrow.*
4. *For all diagrams (0H1V) where A is a discrete valuation ring and where the image of $\mathrm{Spec}(K) \to X$ is a point of codimension 0 on X there is at most one dotted arrow.*

Proof. We have (1) $\Rightarrow$ (2) by Morphisms of Spaces, Lemma 03KU. The implications (2) $\Rightarrow$ (3) and (3) $\Rightarrow$ (4) are immediate. It remains to show (4) implies (1).

Assume (4). We have to show that the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. We already know Δ is representable, separated, a monomorphism, and locally of finite type, see Morphisms of Spaces, Lemma 03HK. Choose an affine scheme U and an étale morphism $U \to X \times_Y X$. Set $V = X \times_{\Delta, X \times_Y X} U$. It suffices to show that $V \to U$ is a closed immersion (Morphisms of Spaces, Lemma 03M4). Since $X \times_Y X$ is locally of finite type over Y we see that U is Noetherian (use Morphisms of Spaces, Lemmas 03XG, 03XH, and 04ZK). Note that V is a scheme as Δ is representable. Also, V is quasi-compact because f is quasi-separated. Hence $V \to U$ is separated and of finite type. Consider a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & V \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & U \end{array}$$

of morphisms of schemes where A is a discrete valuation ring with fraction field K and where K is the residue field of a generic point of the Noetherian scheme V. Since $V \to X$ is étale (as a base change of the étale morphism $U \to X \times_Y X$) we see that the image of $\mathrm{Spec}(K) \to V \to X$ is a point of codimension 0, see Properties of Spaces, Section 04N7. We can interpret the composition $\mathrm{Spec}(A) \to U \to X \times_Y X$ as a pair of morphisms $a, b : \mathrm{Spec}(A) \to X$ agreeing as morphisms into Y and equal when restricted to $\mathrm{Spec}(K)$ and that this restriction maps to a point of codimension 0. Hence our assumption (4) guarantees $a = b$ and we find the dotted arrow in the diagram. By Limits, Lemma 0H1X we conclude that $V \to U$ is proper. In other words, Δ is proper. Since Δ is a monomorphism, we find that Δ is a closed immersion (Étale Morphisms, Lemma 04XV) as desired. $\square$